Project Phase 1: Few-Shot Colorectal Histology Classification

Track A: Research & Development | Group: F26-04

1. Problem Statement

Labelled medical images are scarce and expensive. We study few-shot classification of colorectal tissue images, where a model must recognise tissue types from only a few labelled examples.

Standard ProtoNet builds each class prototype as a plain average of the examples, so one unusual example can ruin it. We are trying to explore if a prototype that adapts to each query, trained with a loss that penalises unusual examples, can beat standard few-shot methods.

2. Dataset

- Kather et al. (2016), colorectal histology, via Kaggle (kmader).
- Link: https://www.kaggle.com/datasets/kmader/colorectal-histology-mnist
- 5,000 colour images (150×150), 8 tissue classes, 625 per class; resized to 224×224.
- Labels are tissue types only. No mutation prediction.

3. Class Split

Eight classes are too few to train and test 5-way on the same dataset. So, we train on other FHIST histology datasets (e.g. CRC-TP, LC25000) and test on Kather's 8 classes, using 5 random classes per episode. As a second check, we also split Kather into 4 training and 4 test classes (4-way).

4. Reference Paper

FHIST: A Benchmark for Few-shot Classification of Histological Images (arXiv:2206.00092).

The paper provides the research background and motivation for applying few-shot learning to histopathology.

https://arxiv.org/abs/2206.00092

5. Method

Dynamic Prototype Adaptation (DPA):

- Weights each support example when building the prototype and adapts the prototype to the query. This still works in 1-shot, where weighting alone cannot.
- Uses a custom loss standard cross-entropy, plus a penalty for giving weight to unusual support examples.
- ViT-Small (DINO pretrained), compared with ResNet-50.

Weighted prototypes already exist, so our contribution is the query-conditioned adaptation combined with the loss that trains it.

6. Experiments

- Settings: 5-way 1-shot, 5-shot and 10-shot.

- Baselines: ProtoNet, MatchingNet, RelationNet, MAML/ANIL, and a fine-tuning baseline.
- Ablations: standard vs. dynamic prototypes; with vs. without our loss; query-based vs. support-only weighting; ViT vs. CNN.
- Evaluation: 600 test episodes, 15 queries per class, 3+ training seeds, same episodes for all methods.
- Metrics: accuracy, precision, recall, macro-F1, with 95% confidence intervals.
- Extra question: why does fine-tuning often match or beat meta-learning, and does our loss help?

7. Scope

In scope: few-shot tissue classification, DPA, baseline comparison and ablations.

Out of scope: mutation prediction, patient diagnosis, treatment advice and clinical use.